

The Nile at NADI 2026 Subtask 1.1: A Two-Phase Acoustic Fine-Tuning Strategy for Robust Dialectal Arabic Speech Recognition

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Abstract

We describe our submission to **NADI 2026 Subtask 1.1 (Noisy Country-Level ASR)** (Sullivan et al., 2026), a two-phase acoustic fine-tuning recipe on WHISPER-LARGE-V3-TURBO. *Phase 1* adapts the model to $\sim 100k$ dialectal utterances (~ 496 h) under stochastic degradation (noise, bandwidth, codec, gain, speed) applied to otherwise clean audio. *Phase 2* anneals on the 22,608-utterance in-domain NADI target at a reduced learning rate. On the official 4,000-utterance blind test set we reach **57.0 WER** (6th of 9); compared with a single-phase model trained on the identical pooled data, the two-phase schedule yields an **8.0 absolute WER point** improvement.

1 Introduction

Whisper (Radford et al., 2023) achieves strong zero-shot performance on Modern Standard Arabic but degrades markedly on Dialectal Arabic (DA) (Hussein et al., 2022; Chowdhury et al., 2021), whose Egyptian, Levantine, Gulf, Iraqi, and Maghrebi varieties differ substantially in phonology and lexicon (Shon et al., 2020). Real-world DA audio adds narrowband encoding, codec artifacts, and background noise—the regime targeted by **NADI 2026 Subtask 1.1** (Sullivan et al., 2026), which ranks systems by WER on utterances from eight countries under degraded conditions.

Naïvely fine-tuning a large model on one dialectal corpus overfits its acoustics without expanding phonetic coverage; conversely, fine-tuning purely on augmented multi-dialect data misaligns outputs from the evaluation distribution (punctuation, orthography, target vocabulary). We propose a **two-phase acoustic fine-tuning** recipe that separates these objectives: **Phase 1** adapts to broad dialects

and acoustic conditions using clean multi-dialect audio under heavy *synthetic* degradation, exposing the model to a diverse span of channel and noise conditions; **Phase 2** anneals on the naturally noisy, in-domain NADI target at a lower learning rate, aligning outputs to the *real* target noise and orthographic distributions. Our contributions are (i) a competitive DA-ASR pipeline on Whisper-Large-v3-Turbo (6th/9 on NADI 2026 Subtask 1.1), (ii) a simple orthogonal degradation pipeline, and (iii) a controlled comparison showing an 8.0 WER-point improvement of the two-phase schedule over a single-phase pooled-data baseline.

2 Related Work

Recent multilingual encoder-decoder speech models—Whisper (Radford et al., 2023), USM (Zhang et al., 2023), SeamlessM4T (Barrault et al., 2023), MMS (Pratap et al., 2024)—are standard starting points for low-resource ASR fine-tuning; Whisper’s LARGE-V3-TURBO halves parameter count by pruning the decoder from 32 to 4 layers while keeping the full 32-layer encoder. Self-supervised alternatives include wav2vec 2.0 (Baeovski et al., 2020), HuBERT (Hsu et al., 2021), XLS-R (Babu et al., 2022), and Conformer encoders (Gulati et al., 2020); see Prabhavalkar et al. (2024) for a survey. Arabic-specific work has progressed via end-to-end and modular systems (Hussein et al., 2022) and unified multilingual code-switching strategies (Chowdhury et al., 2021), with resources like ADI17 (Shon et al., 2020), Common Voice (Ardila et al., 2020), and FLEURS (Conneau et al., 2023) broadening evaluation. Our submission targets NADI 2026 (Sullivan et al., 2026), whose official baseline is zero-shot Whisper-Large-

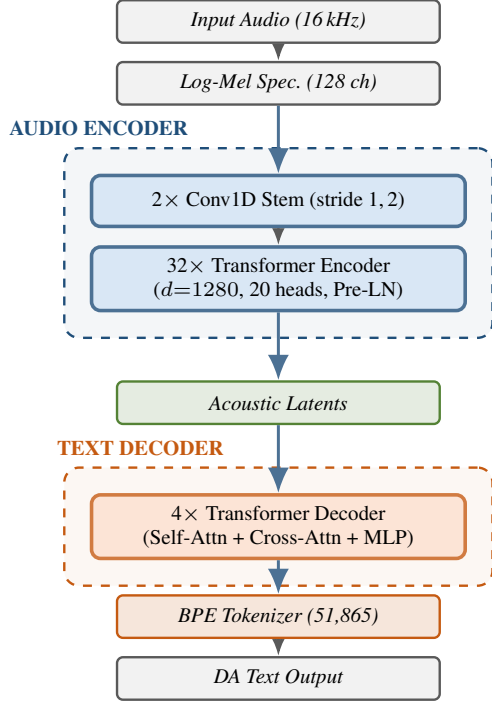


Figure 1: WHISPER-LARGE-V3-TURBO: 32-layer encoder retained, decoder pruned to 4 layers.

v3.

3 Method

Base model. We build on WHISPER-LARGE-V3-TURBO (Radford et al., 2023) ($\sim 809\text{M}$ params). Audio is resampled to 16 kHz mono and converted to a **128-channel** log-Mel spectrogram (25 ms window, 10 ms hop). A two-layer 1D convolutional stem (kernel 3, strides 1 and 2) downsamples the time dimension by a **factor of 2** and projects into $d_{\text{model}} = 1280$. The 32-layer encoder uses Pre-LayerNorm and 20-head self-attention; the 4-layer decoder attends autoregressively over a 51,865-token BPE vocabulary. Figure 1 summarizes the architecture.

Two-phase fine-tuning. **Phase 1** fine-tunes on a $\sim 100\text{k}$ -utterance subset of **ODAMIX** (dialectal-arabic-lahgtna-v2 (oddadmixon, 2026)), a multi-dialect corpus spanning 13 country-level varieties (Maghrebi 33.5%, Egyptian 27.1%, Gulf 14.9%, Levantine 10.9%, Sudanese/Yemeni/Iraqi 13.7%; no MSA). ODA MIX audio is comparatively clean; each mini-batch is passed through the stochastic degradation pipeline of Table 1 to expose the encoder to a synthetic diversity of channel and noise conditions. **Phase 2** anneals the resulting checkpoint on

Transform	Exact setting	P
White noise	Gaussian, SNR 3–15 dB	0.10
Pink noise	$1/f$ spectrum, SNR 3–15 dB	0.10
Bandwidth cut	8 kHz decimation $\rightarrow$ 16 kHz	0.10
Codec sim.	GSM 06.10 full-rate, 13 kbps	0.10
Gain shift	attenuation, -15 to 0 dB	0.10
Speed perturb.	resample, $\{0.9, 1.0, 1.1\}$	0.10

Table 1: Phase-1 stochastic degradation pipeline (independent per example).

the NADI 2026 target at 2×10^{-5} . Unlike Phase 1, the NADI Subtask 1.1 audio is naturally noisy—the subtask is titled “Noisy Country-Level ASR”—so no additional synthetic degradation is applied here; the reduced step size lets the model align to the target’s real noise distribution and output conventions (punctuation, orthography, vocabulary) while limiting catastrophic forgetting of dialectal representations from Phase 1. Figure 2 summarizes the pipeline.

Augmentation pipeline. Six transformations are applied independently at $P=0.10$ each (Table 1), yielding $1 - 0.9^6 \approx 46.9\%$ of examples receiving at least one perturbation. Ranges target orthogonal failure modes—additive noise, narrowband telephony, codec artifacts, quiet/distant speakers, and varied speaking rates—without inducing phonetic drift.

4 Experimental Setup

Data. NADI 2026 Subtask 1.1 (Sullivan et al., 2026) releases 12,800 train + 10,808 validation + 4,000 blind-test utterances over eight countries (Algeria, Egypt, Jordan, Mauritania, Morocco, Palestine, UAE, Yemen). We train on 22,608 (train + 9,808 val) and hold out 1,000 val utterances as internal dev. Phase-1 ODA MIX is a $\sim 100\text{k}$ subset (≈ 496 h) of the 611,180-utterance oddadmixon/dialectal-arabic-lahgtna-v2 (oddadmixon, 2026); the corpus carries no speaker annotations.

Training and decoding. Both phases perform **full fine-tuning** (no frozen layers, no LoRA) with AdamW ($\beta_1=0.9$, $\beta_2=0.999$, $\epsilon=10^{-8}$) on Hugging Face Transformers (Wolf et al., 2020). Table 2 summarizes hyperparameters. Phase 1 ran on $1 \times \text{A100 } 40\text{ GB}$ (≈ 6 h); Phase 2 on $1 \times \text{RTX PRO } 6000$ Blackwell (1 h 20 m, 1,416 steps) with bf16 autocast + fp32 master weights. Decoding uses **greedy search** (beam 1),



Figure 2: Two-phase fine-tuning pipeline. Phase 1 broadens dialectal and acoustic coverage under stochastic degradation on $\sim 100\text{k}$ ODAMIX utterances; the checkpoint is annealed on the NADI 2026 target at a reduced LR in Phase 2.

Parameter	Phase 1	Phase 2
Init. weights	Turbo	Phase-1 ckpt
Dataset	ODAMIX ($\sim 100\text{k}$)	NADI (22,608)
Epochs	3	4
Learning rate	5×10^{-5}	2×10^{-5}
Batch size	32	32
Warm-up	not logged	300 steps
Weight decay	not logged	0.0
Seed	not logged	42
Precision	not logged	bf16 / fp32
GPU	$1 \times \text{A100 40 GB}$	$1 \times \text{RTX 6000}$
Train time	$\approx 6\text{ h}$	1 h 20 m
Max audio / tok	30 s / 448	30 s / 448
Augmentation	$P=0.10$ each	none

Table 2: Hyperparameters. Both phases fine-tune all parameters, greedy decoding. “Not logged” indicates hyperparameters we did not record for the Phase-1 run.

language="ar", task="transcribe", max 225 output tokens.

Evaluation. We report WER on our 1,000-utterance internal dev (in-loop scoring: diacritic + tatweel stripping) and the official 4,000-utterance test set. **All test-set WER values reported in Table 3 are computed with the official organizers’ scorer** (which additionally normalizes digits, punctuation, and Alif/Ya/Ta-Marbouta variants). Dev figures are therefore slightly pessimistic relative to the official protocol.

5 Results

Table 3 reports WER. Zero-shot Whisper-Turbo reaches 80.0; direct target fine-tuning reduces to 71.5; Phase 1 alone reaches 65.0, confirming that broad multi-dialect exposure under degradation transfers even without target supervision. The full two-phase pipeline reaches **57.0**—a 23.0-point absolute (28.8% relative) reduction over zero-shot, **ranking 6th of 9** submitted systems.

Two-phase vs. single-phase pooled baseline. Pooling the identical Phase-1 and target corpora into a single training run reaches only 65.0—matching Phase 1 alone—while the two-phase split reaches 57.0. *Compared with the single-phase*

pooled-data baseline, the two-phase schedule improves WER by 8.0 absolute points (12.3% relative), holding the training corpora fixed. We refrain from a stronger causal claim (“schedule alone”) because the two configurations differ modestly in effective LR schedule and total step count (Phase 1: 3 epochs at 5×10^{-5} on $\sim 100\text{k}$ utts, then Phase 2: 4 epochs at 2×10^{-5} on 22.6k utts; single-phase pooled: comparable epochs at a single LR on the union). We interpret the finding as: given a fixed corpus, splitting fine-tuning into a high-LR broad-adaptation stage and a low-LR domain-annealing stage substantially outperforms a single flat schedule.

Why the decomposition helps. A single-phase target fine-tune memorizes target acoustics without learning dialect-general representations; a single-phase ODAMIX fine-tune produces a robust but stylistically misaligned model. The two-phase recipe assigns each objective to the phase best suited to optimize it: broad synthetically-degraded data at higher LR for representation learning, real in-domain data at lower LR for output alignment.

6 Conclusion

We described our NADI 2026 Subtask 1.1 submission: a two-phase acoustic fine-tuning strategy on Whisper-Large-v3-Turbo that separates *dialectal/acoustic robustness* (Phase 1, synthetic degradation on clean multi-dialect data) from *target-domain alignment* (Phase 2, annealing on the naturally noisy in-domain NADI target). The pipeline achieves 57.0 WER on the official blind test set (6th/9), an 8.0-point improvement over a single-phase pooled-data baseline trained on the identical corpora.

Limitations

Several caveats bound our claims. First, we compare the two-phase schedule against single-phase pooled training, but we do not isolate augmentation itself: a Phase 1 on clean ODAMIX *without* synthetic degradation, followed by annealing,

System	Dev	Test	Δ^a	Rel.% ^b
Whisper-Turbo (zero-shot)	—	80.0	—	—
+ Phase 2 only (target FT)	—	71.5	−8.5	10.6%
+ Phase 1 only (ODAMIX+aug.)	—	65.0	−15.0	18.8%
+ Phase 1 → Phase 2 (ours, rank 6/9)	61.4	57.0	−23.0	28.8%
<i>Ablation — same data, no two-phase schedule (official test)</i>				
Combined single-phase (no annealing)	—	65.0	−15.0	18.8%

Table 3: WER (%) on NADI 2026 Subtask 1.1; all test values computed with the official organizers’ scorer. ^aAbsolute WER reduction vs. zero-shot. ^bRelative reduction vs. zero-shot. Our two-phase pipeline (highlighted) ranks **6th of 9 submitted systems**. Splitting the same pooled data into two phases yields an **8.0-point improvement** over the single-phase baseline (65.0 → 57.0), a **12.3% relative** reduction under identical training corpora.

would be required to separate the effect of augmentation from that of the additional dialectal data—we leave this to future work. Second, the two-phase vs. single-phase comparison does not perfectly match total step count or LR schedule, so the 8.0-point improvement should be read as a comparison against a natural single-phase baseline rather than a claim about the schedule “alone”. Third, only Phase 2 was run with a fixed seed (42); Phase-1 seed, warm-up, precision, and weight decay were not logged, and we report no variance across seeds in either phase. Fourth, ODAMIX has no speaker identifiers, so speaker overlap between phases cannot be audited. Fifth, the Phase 1 corpus is Maghrebi/Egyptian-heavy with no MSA; performance on under-represented dialects (e.g. Iraqi, 2.1% of P1) may be systematically weaker than the aggregate WER suggests. Finally, reverberation and room-impulse-response convolution are not part of our augmentation set; both are natural extensions.

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